MELUSI SIBANDA

Software Developer · Database Engineer · Full-Stack Builder

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PROFESSIONAL SUMMARY

Results-driven Software Developer and ICT graduate (Advanced Diploma, University of Mpumalanga, 2025) with production experience designing enterprise-grade database architecture at Nqola Logistics. Deep expertise in SQL/PostgreSQL, RESTful API development, and full-stack web technologies. Proven ability to reduce reporting overhead by 60 %+ through automated pipelines and optimised query design. Equally comfortable building React frontends, backend APIs, and robust data models — and committed to clean, maintainable code that scales.

CORE COMPETENCIES

✦ PostgreSQL & SQL	✦ Database Schema Design	✦ Query Optimisation
✦ REST API Development	✦ Node.js / Django / Jakarta EE	✦ React & TypeScript
✦ Python · Java · C# · PHP	✦ Git & CI/CD Pipelines	✦ Containers & Kubernetes
✦ Data Modelling & Integrity	✦ Agile / Scrum	✦ Cybersecurity Fundamentals

PROFESSIONAL EXPERIENCE

Software Developer Intern

Nqola Logistics — Johannesburg | Jan 2026 – Present

- Architected a normalised relational database schema for an enterprise fleet management system — vehicles, drivers, routes, and service records — enforcing referential integrity constraints across all modules with zero data corruption.
- Wrote and optimised complex SQL queries powering real-time dashboard reporting; applied strategic indexing to cut execution time on large vehicle and trip datasets.
- Built a RESTful API backend integrating live GPS telemetry with a PostgreSQL data layer, delivering real-time visibility across 100 % of active fleet assets.
- Managed database migrations and schema versioning via Git, maintaining a clean, auditable change history throughout the development lifecycle.
- Implemented data-validation and error-handling routines to guarantee data quality in a high-volume operational environment.
- Reduced manual reporting time by ~60 % through automated data pipelines and structured database reporting workflows.

KEY PROJECTS

iCurate — Hospital Administration System

- Designed a fully normalised relational database for patient records, appointments, and clinical workflows.
- Implemented Role-Based Access Control (RBAC) with granular data-access policies across user roles.

AgriNathi — Smart Farming Platform

- Built a data-driven backend with structured modelling for crop records, resource inputs, and weather API integration.
- Applied strict data-consistency principles throughout ingestion and reporting layers.

Bug Tracking System

- Developed a full-stack web platform with a structured SQL backend for logging, categorising, and tracking software defects.

- Delivered query-driven reporting dashboards giving team leads real-time defect visibility.

Egumeni Eats · Your Look · Mpumalanga Tourism (Portfolio)

- Delivered responsive React / TypeScript frontends with animated UI, multi-step forms, and live API integration.

EDUCATION

Advanced Diploma in IT (Application Development)
University of Mpumalanga · 2025

Diploma in Information Technology
University of Mpumalanga · 2022 – 2024

National Senior Certificate
Highlands North Boys High / Star Schools Braamfontein · 2015 / 2021

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Microsoft SC-900: Security, Compliance & Identity Fundamentals** — Microsoft · 2025
- **Microsoft 365 Certified: Fundamentals** — Microsoft · 2025
- **Microsoft AI Fluency** — Microsoft · 2025
- **Full Stack Developer Certificate** — FNB App Academy · 2025
- **Introduction to Containers, Kubernetes & OpenShift V2** — IBM SkillsBuild · 2022
- **SAP FI Training** — Mentec Foundation · 2024
- **Mpumalanga AI Hackathon — Scrum Master** — 2025
- **Mpumalanga AI Hackathon** — 2023

References available upon request · melusi-sibanda-portfolio.onrender.com